Exercise sheet n°1

In this session, we consider online learning with experts (see Lecture #1) with linear losses. The losses ℓ_{jt} are in $[0,1]$ when not precised otherwise.

Exercise 1 :

Consider online learning with experts (see Lecture #1) with linear losses. Show that no strategy satisfies for all sequence $(\ell_{1t},\dots,\ell_{Nt})_t \in ([0,1]^N)^{\mathbb{N}}$:

$$\sum_{t=1}^{T}\sum_{j=1}^{N} p_{jt}\ell_{jt} - \sum_{t=1}^{T}\min_{k\in[N]} \ell_{kt} = o(T).$$

Exercise 2 :

Consider online learning with experts (see Lecture #1) with linear losses. Assume in this exercise that $\ell_{jt} \in [m,M]$, with $m, M \in \mathbb{R}$ unknown. How can we tune η?
We consider in the following EWA with adaptive rates $(\eta_t)_t$:

$$p_{jt} = \frac{e^{-\eta_t \sum_{s=1}^{t-1}\ell_{js}}}{\sum_{k=1}^{N} e^{-\eta_t \sum_{s=1}^{t-1}\ell_{ks}}}.$$

1) Show that if (η_t) are non-increasing

$$\frac{1}{N}\sum_{j=1}^{N} p_{jt}e^{-\eta_t \ell_{jt}} \geq \frac{1}{N^{\frac{\eta_t}{\eta_{t+1}}}} \frac{\left(\sum_{j=1}^{N}\exp(-\eta_{t+1}\sum_{s=1}^{t}\ell_{js})\right)^{\frac{\eta_t}{\eta_{t+1}}}}{\sum_{k=1}^{N}\exp\left(-\eta_t\sum_{s=1}^{t-1}\ell_{ks}\right)}.$$

Hint: Use the fact that $x \mapsto x^{\frac{\eta_t}{\eta_{t+1}}}$ is convex.

2) Show that if (η_t) are non-increasing, then the regret of EWA satisfies:

$$R_T \leq \frac{\ln N}{\eta_T} + \sum_{t=1}^{T}\delta_t,$$

where $\delta_t = \sum_{j=1}^{N} p_{jt}\ell_{jt} + \frac{1}{\eta_t}\ln\left(\sum_{j=1}^{N} p_{jt}e^{-\eta_t\ell_{jt}}\right)$.

Hint: Multiply by $\frac{-1}{\eta_t}$ the logarithm of the expression obtained in **1)** to make a telescopic sum appears.

Recall the Bernstein's inequality for a random variable $X \in [m,M]$:

$$\forall \eta > 0,\ \ln \mathbb{E}[e^{\eta X}] \leq \eta\mathbb{E}[X] + \frac{e^{\eta(M-m)} - 1 - \eta(M-m)}{(M-m)^2}\mathrm{Var}\,(X)\,.$$

We now consider EWA with $\eta_t = \frac{\ln N}{\sum_{s=1}^{t-1}\delta_s}$, with the convention that $\frac{\ln N}{0} = +\infty$.

3) Let $v_t = \sum_{j=1}^{N}(\ell_{jt} - \sum_{k=1}^{N} p_{kt}\ell_{kt})^2 p_{jt}$.

(a) Show that $v_t \geq \dfrac{\eta_t(M-m)}{e^{\eta_t(M-m)} - \eta_t(M-m) - 1}(M-m)\delta_t$.

(b) Deduce that $v_t \geq \frac{2\delta_t}{\eta_t} - \frac{2}{3}(M-m)\delta_t$.

4)

(a) Show that $\left(\sum_{t=1}^{T} \delta_t\right)^2 \leq \sum_{t=1}^{T} v_t \ln N + (M-m)(1 + \frac{2}{3}\ln N)\sum_{t=1}^{T}\delta_t$.

(b) Finally, show that $R_T \leq (M-m)\sqrt{T\ln N} + (M-m)(2 + \frac{4}{3}\ln N)$.

Exercise 3 :

Consider the ε-greedy algorithm with $\varepsilon_t = \min\left(1, \frac{(K\ln(t))^{\frac{1}{3}}}{t^{\frac{1}{3}}}\right)$ for any $t \in \mathbb{N}$. Show that for a large enough universal constant $C > 0$, the regret of ε-greedy satisfies

$$R_T \leq CT^{\frac{2}{3}}(K\ln(T))^{\frac{1}{3}}.$$

Hint: Bound the instantaneous regret $\mathbb{E}[\Delta_{a_t}]$.

Exercise 4 :

Concentration for sequences of random length. Let $X_1, X_2, \ldots$ be a sequence of independent standard Gaussian random variables defined on probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Suppose that $T : \Omega \to \{1, 2, 3, \ldots\}$ is another variable and let $\hat{\mu}_T = \sum_{t=1}^{T} \frac{X_t}{T}$ be the empirical mean based on T samples.

1) Show that if T is independent from X_t for all t, then for any $\delta \in (0,1)$

$$\mathbb{P}\left(\hat{\mu}_T \geq \sqrt{\frac{2\ln(1/\delta)}{T}}\right) \leq \delta.$$

2) Now relax the assumption that T is independent from $(X_t)_t$. Let $E_t = \mathbb{1}_{T=t}$ and $\mathcal{F}_t = \sigma(X_1, \ldots, X_t)$ be the σ-algebra generated by the first t samples. Let $\delta \in (0,1)$ and show there exists a random variable T such that for all t, E_t is $\mathcal{F}_t$-measurable and

$$\mathbb{P}\left(\hat{\mu}_T \geq \sqrt{\frac{2\ln(1/\delta)}{T}}\right) = 1.$$

Hint: You can use the law of the iterated logarithm, which says if $X_1, X_2, \ldots$ is a sequence of independent and identically distributed random variables with zero mean and unit variance, then

$$\limsup_{n\to\infty} \frac{\sum_{t=1}^{n} X_t}{\sqrt{2n\ln\ln n}} = 1 \quad \text{almost surely.}$$

3) What is the relation between the above inequality and our concentration lemma for the empirical means in bandits problems? Do **2)** and our lemma contradict? Why?

Exercise 5 :

Phased SE. Consider the following phased Successive Eliminations algorithm parameterized by $a > 1$.

Algorithm: Phased Successive Eliminations
input: T, $a \geq 1$
$\mathcal{K} \leftarrow [K]$
$\ell \leftarrow 0$
while $\mathrm{Card}(\mathcal{K}) > 1$ **do**
  pull all arms in $\mathcal{K}$ $\lceil a^\ell \rceil$ times
  for *all* $k \in \mathcal{K}$ *such that* $\hat{\mu}_k + \sqrt{\frac{2\ln T}{N_k(T)}} \leq \max_{i\in\mathcal{K}} \hat{\mu}_i - \sqrt{\frac{2\ln T}{N_i(T)}}$ **do** $\mathcal{K} \leftarrow \mathcal{K} \setminus \{k\}$
  $\ell \leftarrow \ell + 1$
repeat pull only arm in $\mathcal{K}$ **until** $t = T$

1) Show a regret bound similar to Successive Eliminations algorithm.

2) What is the role played by a?

Exercise 6 :

Adapting to reward variance. Let $X_1, \ldots, X_N$ be a sequence of i.i.d. random variables with mean μ, variance σ^2 and bounded support so that $X_t \in [0, M]$ almost surely. Define the estimators

$$\hat{\mu}_N = \frac{1}{N}\sum_{t=1}^{N} X_t$$

$$\hat{\sigma}^2 = \frac{1}{N}\sum_{t=1}^{N}(\hat{\mu} - X_t)^2.$$

We admit in the following the **empirical Bernstein** inequality:

$$\mathbb{P}\left(|\hat{\mu}_N - \mu| \geq \sqrt{\frac{2\hat{\sigma}^2}{N}\ln(3/\delta)} + \frac{3M}{N}\ln(3/\delta)\right) \leq \delta.$$

1) Show that the Bernstein inequality in the course implies here

$$\mathbb{P}\left(|\hat{\mu}_N - \mu| \geq \sqrt{\frac{2\sigma^2}{N}\ln(2/\delta)} + \frac{2M}{3N}\ln(2/\delta)\right) \leq \delta$$

Comment on the differences between the two above *Bernstein inequalities*.

2) Show that $\hat{\sigma}^2 = \frac{1}{N}\sum_{t=1}^{N}(X_t - \mu)^2 - (\hat{\mu}_N - \mu)^2$.

3) Is $\hat{\sigma}^2$ an unbiased estimator of σ^2? If not, can we easily make it unbiased?

4) Show that

$$\mathbb{P}\left(\hat{\sigma}^2 \geq \sigma^2 + \sqrt{\frac{2M^2\sigma^2}{N}\ln(1/\delta)} + \frac{2M^2}{3N}\ln(1/\delta)\right) \leq \delta.$$

Hint: Use Bernstein inequality of **1)**.

5) **(Hard)** Consider a bandit setting with K arms, bounded rewards $X_k(t) \in [0, M]$ and the variance of the k-th arm is σ_k^2. Design a policy that depends on M, but does not need to know σ_i a priori, such that there exists a universal constant $C > 0$ with

$$R_T \leq C \sum_{k,\Delta_k>0} \left(\Delta_k + (M + \frac{\sigma_i^2}{\Delta_i}) \ln T\right).$$

Hint: Without the stack of rewards model, the empirical Bernstein inequality can be extended (up to some changes) to cases where N is a random variable.

Exercise 7 :

This exercise studies the celebrated Thompson sampling algorithm, described below.

In words, Thompson sampling starts with a prior distribution $\boldsymbol{p}_0$ distribution on the (mean) parameters of the bandits instance and at each round t, it draws random samples $\theta_k(t)$ from the posterior distribution $\boldsymbol{p}_{t-1}$ on the instance parameters at time $t-1$, which is defined as

$$\boldsymbol{p}_{t-1}(A) = \mathbb{P}\Big((\mu_1, \ldots, \mu_K) \in A \mid \mathcal{F}_{t-1}\Big) \quad \text{for any } A \in B(\mathbb{R}), \tag{1}$$

where $\mathcal{F}_{t-1} = \sigma\Big(U_1, X_{a_1}(1), U_2, X_{a_2}(2), \ldots X_{a_{t-1}}(t-1)\Big)$ and the U_s are random variables uniformly drawn in $[0, 1]$, that are independent with all other variables.

Algorithm: Thompson sampling
input: prior distribution $\boldsymbol{p}_0$
for $t = 1, \ldots, T$ **do**
  Sample $\boldsymbol{\theta}(t) \sim \boldsymbol{p}_{t-1}$
  Pull $a_t \in \operatorname{argmax}_{k\in[K]} \theta_k(t)$ `// Ties broken arbitrarily`
  Update $\boldsymbol{p}_t$ as the posterior distribution of the parameters, following Bayes rule.

We note for each time $t \in \mathbb{N}$ and arm $k \in [K]$:

$$S_k(t) = \sum_{s=1}^{t} X_k(s)\mathbb{1}_{a_s=k}.$$

1) Consider an instance of Bernoulli bandits, i.e., $\mathcal{D} = \{\text{Bernoulli}(\mu) \mid \mu \in [0, 1]\}^K$. Show then that in the case of Bernoulli rewards with a uniform prior, at each time $t \in \mathbb{N}$, $\boldsymbol{p}_{t-1}$ is the joint distribut1ion of K independent Beta distributions, where the k-th Beta distribution has

parameters $(S_k(t-1)+1, N_k(t-1)-S_k(t-1)+1)$. In other words for any $t \in \mathbb{N}$, the drawn samples $\theta_k(t)$ are independent with each other conditioned on $\mathcal{F}_{t-1}$ and

$$\theta_k(t) \sim \mathrm{Beta}(S_k(t-1)+1, N_k(t-1)-S_k(t-1)+1).$$

2) Consider now that the prior is the improper uniform distribution[1] on $\mathbb{R}$ and Gaussian bandits with variance σ^2, i.e., $\mathcal{D} = \{\mathcal{N}(\mu, \sigma^2) \mid \mu \in \mathbb{R}\}^K$.
For any $t \in \mathbb{N}$, what is the distribution of $\boldsymbol{p}_{t-1}$ in this case?

[1] This can be seen as the uniform distribution on $\mathbb{R}$. It is not a proper distribution, since it is not of measure 1, but the Bayes rule can still be applied with it.